

NICOLE HUIMEI LEE

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EDUCATION

Massachusetts Institute of Technology Sep 2025 – Present
S.M./Ph.D., Aeronautical and Astronautical Engineering | Dual S.M., Technology and Policy
2026-2027 President of Graduate Association of Aeronautics and Astronautics

California Institute of Technology Sep 2021 – Jun 2025
B.S., Double major in Mechanical Engineering and Business, Economics, and Management

RESEARCH EXPERIENCE

MIT Human Systems Laboratory Sep 2025 – Present
Graduate Researcher, Prof. Dava Newman Cambridge, MA

- Developing a wearable neuromuscular electrical stimulation (NMES) system as an exercise countermeasure to musculoskeletal deconditioning, including an electromechanical electrode model guiding sizing and placement within a gravity-loading countermeasure suit.
- Designed and executed human-subject protocols quantifying metabolic, cardiovascular, and perceptual response to NMES via indirect calorimetry and heart rate variability.

MIT Media Lab 2025 – Present
Research Contributor Cambridge, MA

- Development of wearable nanoelectronic ultrasound technology for biomedical monitoring and breast cancer applications for MIT Health and Life Sciences Collaborative in Canan Dagdeviren's lab.

Caltech Graduate Aerospace Laboratories (GALCIT) Oct 2021 – Jun 2025
Undergraduate Researcher, Prof. Mory Gharib Pasadena, CA

- Simulated fluid-structure interaction in compliant tubes using coupled lattice Boltzmann and immersed boundary solvers, characterizing cardiovascular flow across pulsatile regimes.
- Applied physics-informed neural networks to fuse sparse experimental and computational aerodynamics data, reconstructing full 3-D flow fields from limited measurements.

PROFESSIONAL EXPERIENCE

The Aerospace Corporation – Center for Space Policy and Strategy May 2026 – Sep 2026
Research Fellow, Matthew Isakowitz Scholar Pentagon City and Washington, DC

- Developed a policy-analysis framework evaluating the ISS-to-commercial LEO transition, synthesizing technical, programmatic, and regulatory considerations for government and industry stakeholders; output is a research paper and policy brief on the Hill.

Space Capital Jun 2024 – Aug 2024
Venture Capital Intern, Brooke Owen Fellow New York, NY

- Built market analysis models for direct-to-cell satellite communications supporting investment evaluation and portfolio due diligence; published the resulting thesis as *Connecting the World*.

The Aerospace Corporation Jun 2023 – Aug 2023
Propulsion Intern El Segundo, CA

- Developed solid motor propellant grain regression tooling and reconstructed ballistic performance from static fire test data, supporting assessments for NASA, the U.S. Space Force, and partner organizations.

Quest Carbon Capture and Storage Jun 2022 – Sep 2022
Geomechanics Intern Alberta, Canada

- Modeled reservoir compaction for a carbon storage site and linked geomechanical simulation tools with Petrel for subsurface analysis.

Boeing Aeronautics Mar 2022 – Jul 2022
Fluid Mechanics Intern Remote

- Developed machine learning models predicting 3-D flow fields over airfoil and speed bump geometries

adopted into airwing computation workflows.

SELECTED ENGINEERING PROJECTS

- NASA RASC-AL Theme 4: Smart / Resilient Lunar Habitat** 2026 – Present
Human Factors Subteam Co-Lead, MIT Team
Leading human factors design for MIT's lunar habitat concept, developing crew operational scenarios, habitability requirements, and emergency response criteria.
- NASA From Mars to Table Challenge** 2025 – 2026
Human Factors Subteam Lead
Led human factors design of a Mars food production system, evaluating crew workflows, ergonomics, maintenance demands, and operational constraints for system requirements.
- Smart Skinsuit Parabolic Flight** Nov 2026
Mission Director, MIT Human Systems Lab
Directing a November 2026 parabolic flight demonstration of MIT's Smart Skinsuit and developed NMES hardware, overseeing mission preparation, experiment integration, and in-flight testing.

FELLOWSHIPS, HONORS & AWARDS

- Payload Pioneers 30 Under 30** 2026
- Matthew Isakowitz Commercial Space Scholarship** 2026
- NSF Graduate Research Fellowship** 2025
- Brooke Owens Fellowship** 2024
- MIT Schwarzman College of Computing SERC Scholar** 2026
- Los Angeles Philanthropic Foundation Scholar**
- Additional Honors & Awards:** ASCENT Fellow (2026); Aerospace Corporation Iron Intern; Beckman Political Award; Kiyo and Eiko Tomiyasu SURF Scholar; LeRoy and Anita Nelson Scholarship; Mr. Tao Tan Scholarship; Peat Scholarship; Arizona Seal of Fine Arts Award
- Leadership:** Co-President, MIT Graduate Association of Aeronautics and Astronautics ; Co-President, MIT Space Industry Club

SELECTED PUBLICATIONS & PRESENTATIONS

- 2027** “Electromechanical Modeling of Wearable NMES Electrodes in a Gravity-Loading Countermeasure Suit,” *IEEE Aerospace Conference* (accepted).
- 2026** “Feasibility of a Wearable Neuromuscular Electrical Stimulation System as an Adjunct Exercise Countermeasure for Long-Duration Human Spaceflight,” *International Conference on Environmental Systems*.
- 2025** “Analysis and Forecast of Direct-to-Cellular Satellite Communication in the Space Economy,” *Global Space Exploration Conference (GLEX)*, New Delhi, India. Interactive presentation and conference paper.
- 2024** “Development of Machine Learning Tools for Aerospace Design: Wind Tunnel Investigations on a Speed Bump Model,” *AIAA SciTech Forum*.
- 2022** “Reservoir Compaction Poroelastic Model,” *Caltech Summer Undergraduate Research Fellowships (SURF)*, research presentation.

TECHNICAL SKILLS

- Programming & ML:** Python, MATLAB, Julia, R, C, Java, JavaScript; PyTorch, TensorFlow, JAX, scikit-learn, NumPy, SciPy, pandas
- Modeling & Simulation:** CFD, FEA, FSI, PINNs, Bayesian inference, Monte Carlo, lattice Boltzmann and immersed boundary methods; ANSYS, COMSOL, Simulink
- Engineering Software:** SOLIDWORKS, AutoCAD, NX, LabVIEW, CAESAR II, Petrel
- Biomedical & Experimental:** NMES, wearable ultrasound, indirect calorimetry, COSMED K5, Polar H10, HRV, physiological sensing, impedance measurement, BLE, human-subject experiments
- Prototyping & Certifications:** 3-D printing, CNC machining, laser cutting, waterjet cutting, Git, LaTeX; certified scuba diver, Caltech Prototyping Laboratory, construction safety training